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HANDSET REV A / SYSTEM SCHEMATIC INDEX

Engineering draft. Retained ESP32-S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces.

A

ESP32 host, boot and reset



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Key matrix and trackball



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USB-C data and ESD



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B

Fused USB input and eFuse



File: usb-input.kicad_sch

USB-C source detection and permission



File: usb-detect.kicad_sch

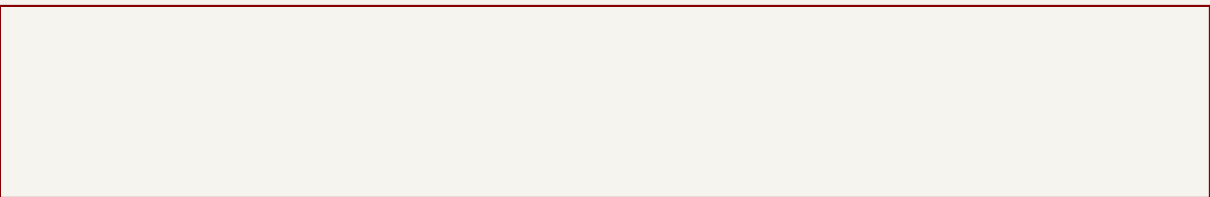
Battery, thermistor and programmable charger



File: power.kicad_sch

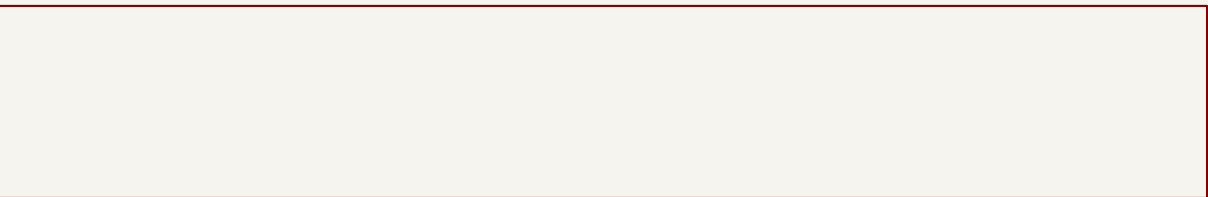
C

Power control and fuel gauge



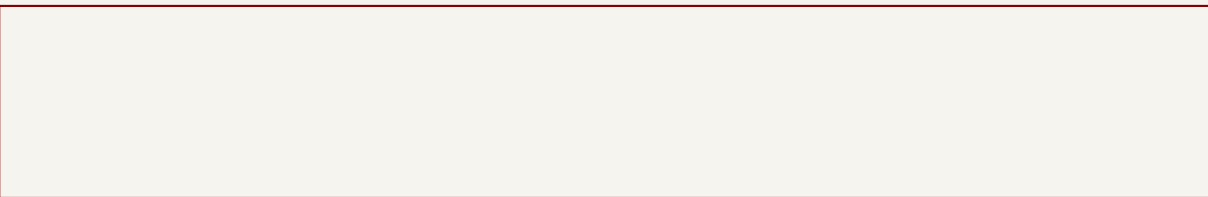
File: power-control.kicad_sch

Main 3.3 V buck-boost supply



File: reg3.kicad_sch

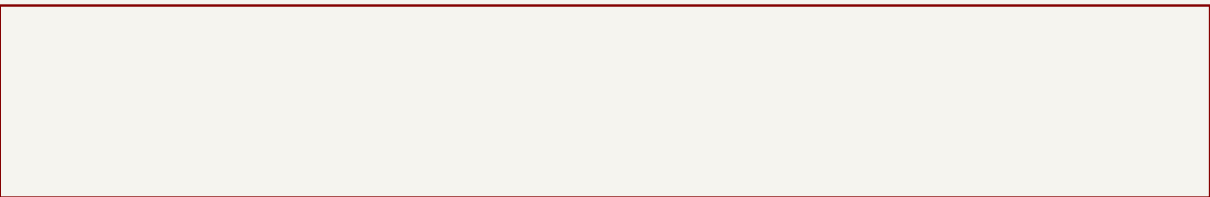
Auxiliary 5 V and 1.8 V supplies



File: aux-power.kicad_sch

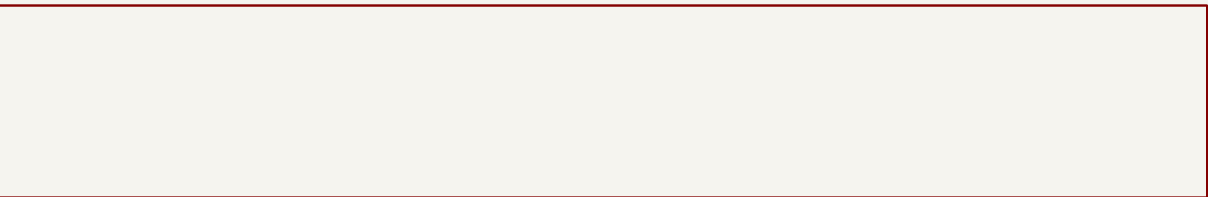
D

Regulated modem supply and enable



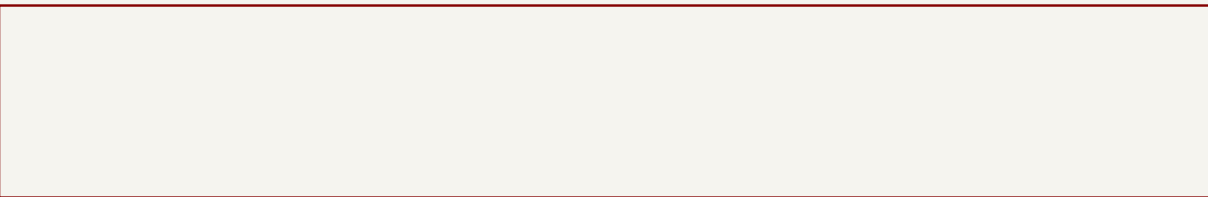
File: modem-power.kicad_sch

SX1262 LoRa radio



File: radios.kicad_sch

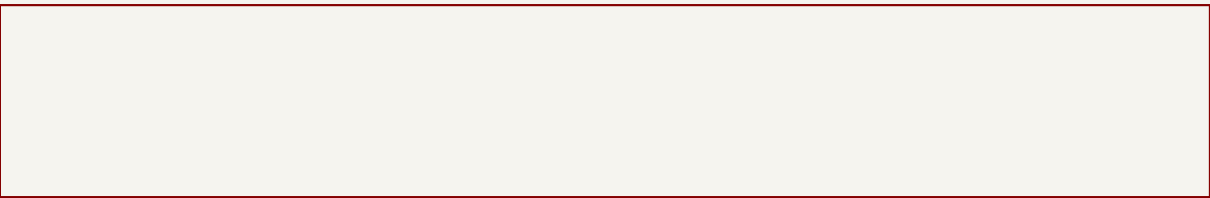
CC1101 sub-GHz radio



File: cc-radio.kicad_sch

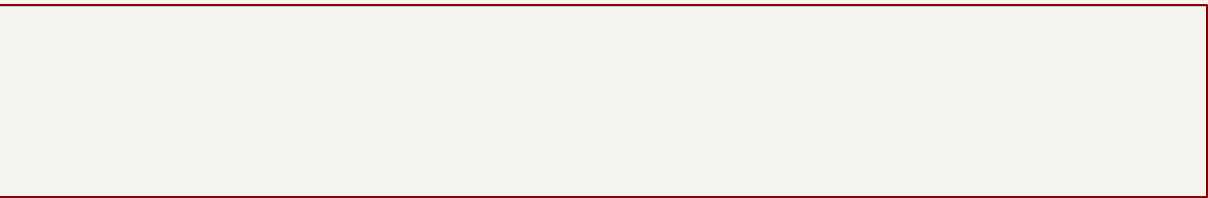
E

125 kHz reader analog circuit



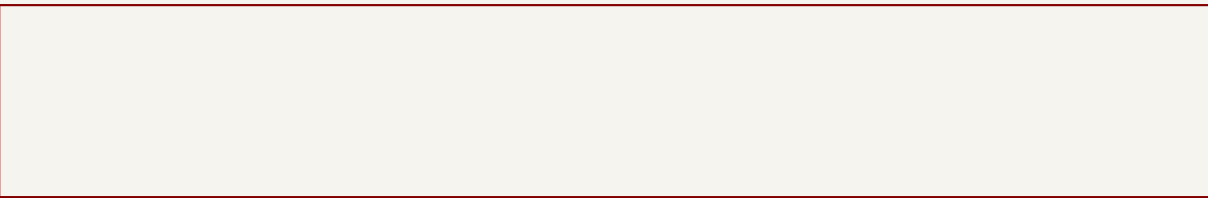
File: lf-radio.kicad_sch

LF reader logic translation



File: lf-support.kicad_sch

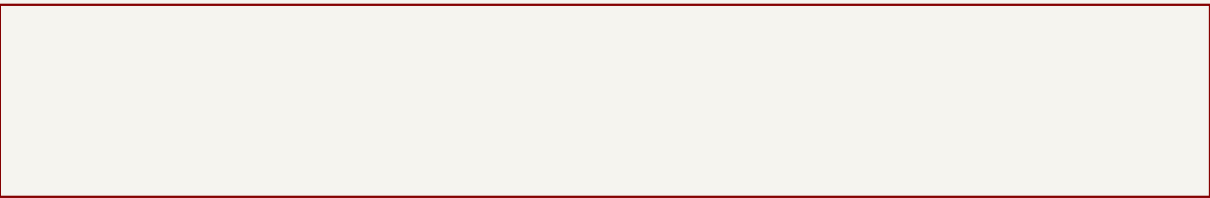
GNSS receiver and passive antenna



File: gnss-antennas.kicad_sch

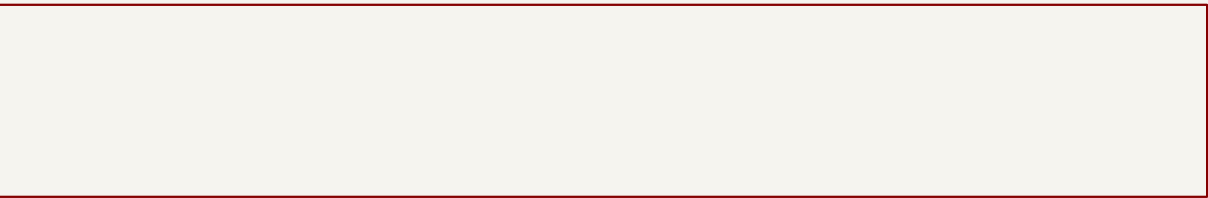
F

NFC reader, oscillator and local supplies



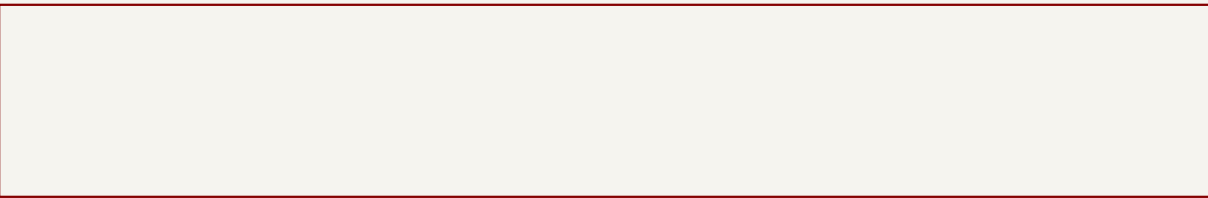
File: nfc.kicad_sch

Media speaker amplifier and digital microphone



File: audio.kicad_sch

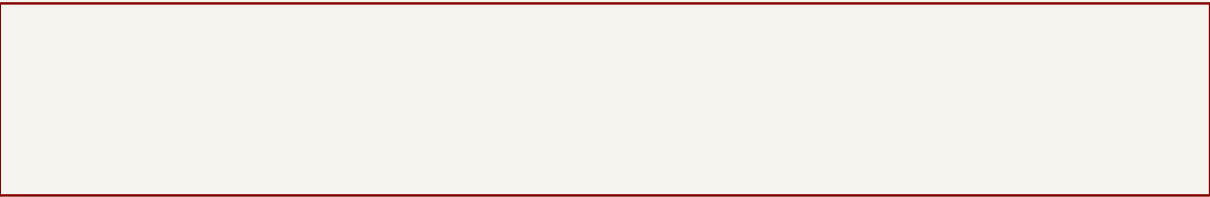
Cellular adapter and call-audio harnesses



File: interfaces.kicad_sch

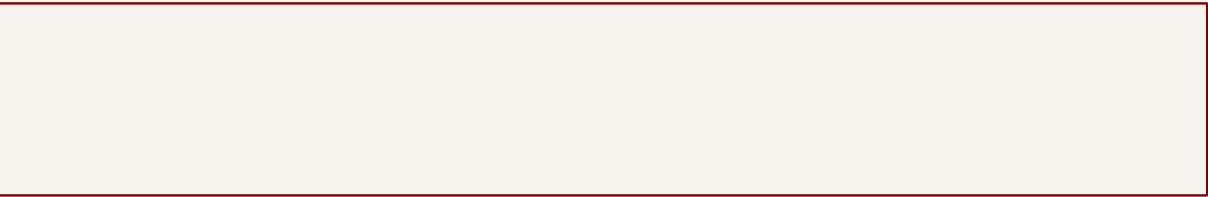
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Protected external module port



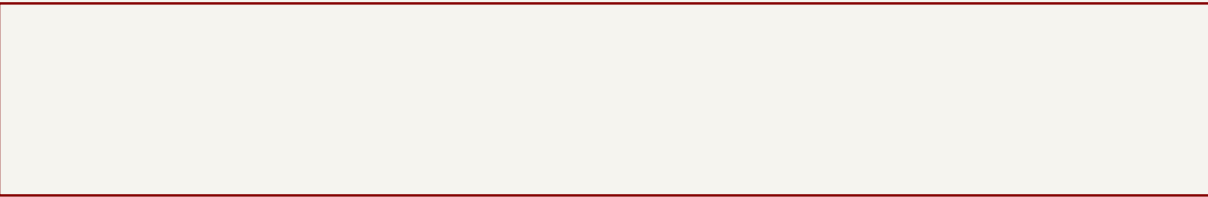
File: protection.kicad_sch

MicroSD and case-mounted Infrared



File: storage-ir.kicad_sch

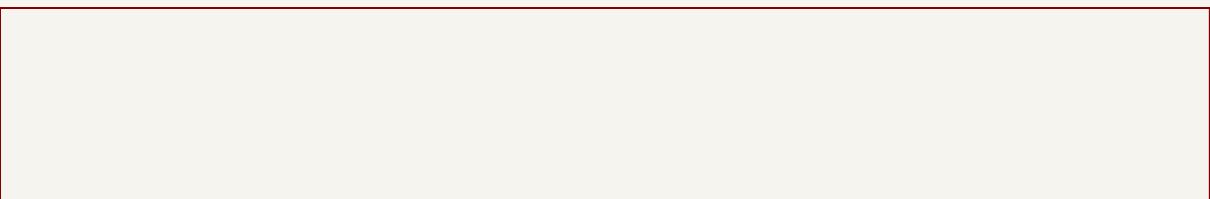
External display, touch and backlight



File: display.kicad_sch

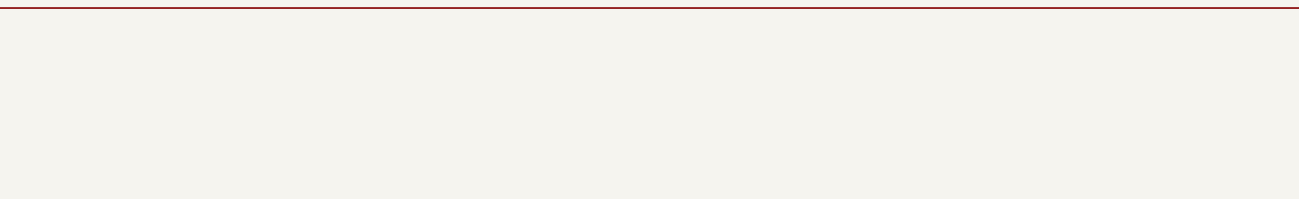
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Always-on charging supervisor



File: power-supervisor.kicad_sch

Display supply and translation

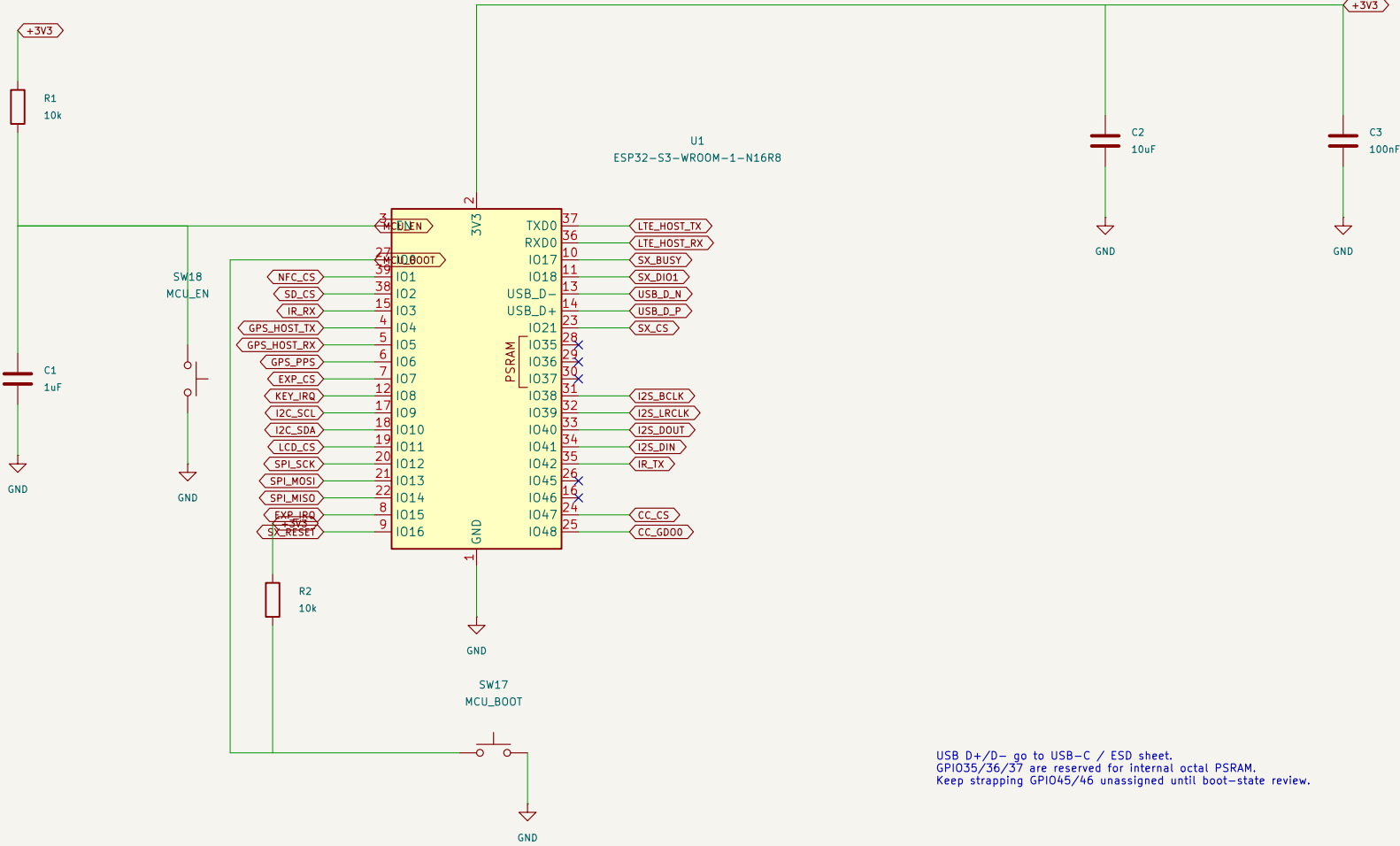


File: display-interface.kicad_sch

Sheet: /		
File: handset.kicad_sch		
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Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6	Id: 1/24	

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

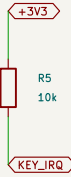
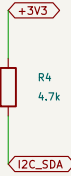
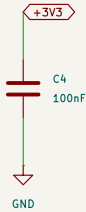
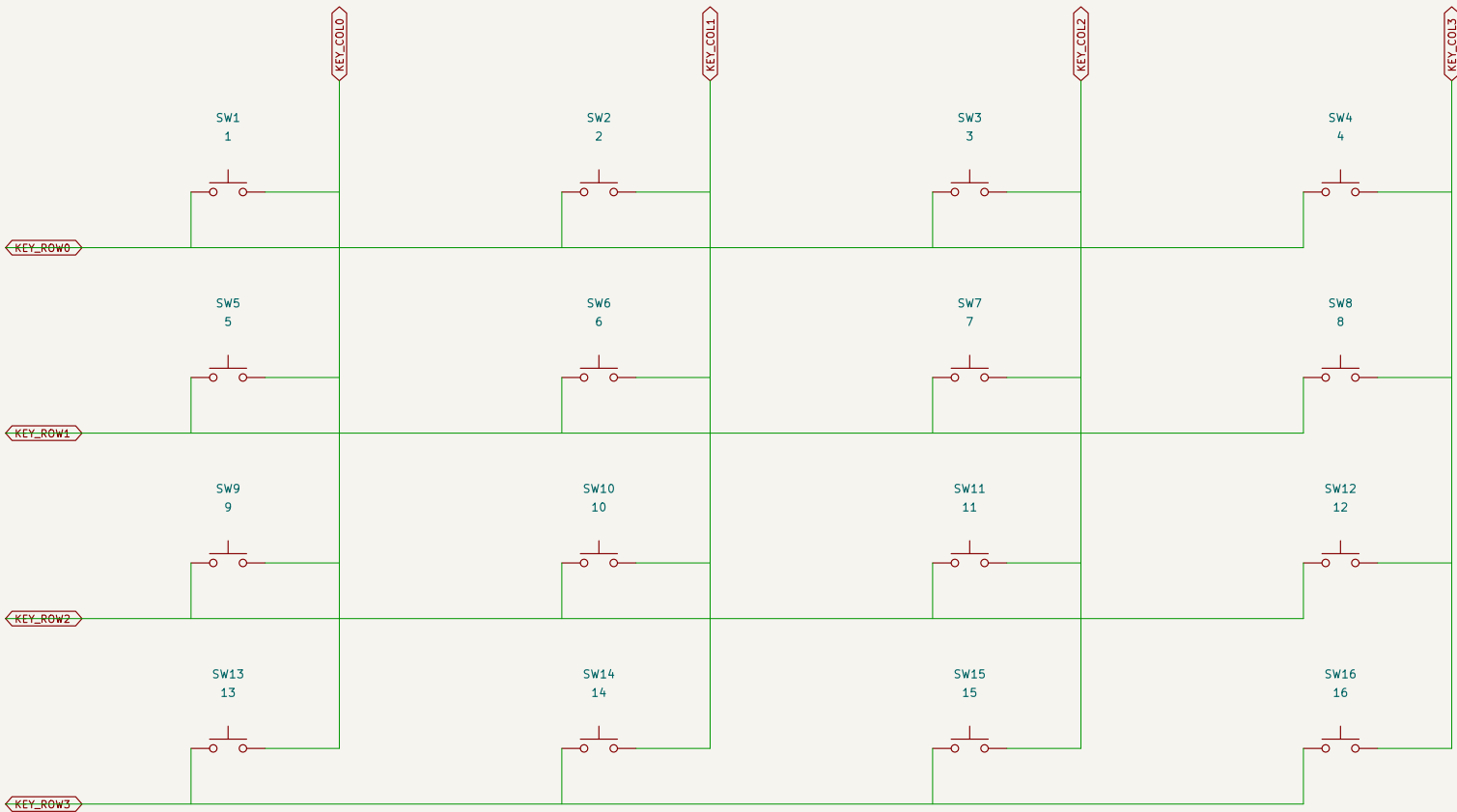
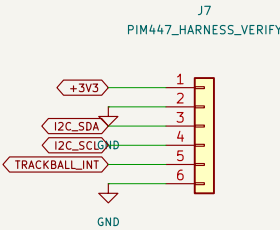
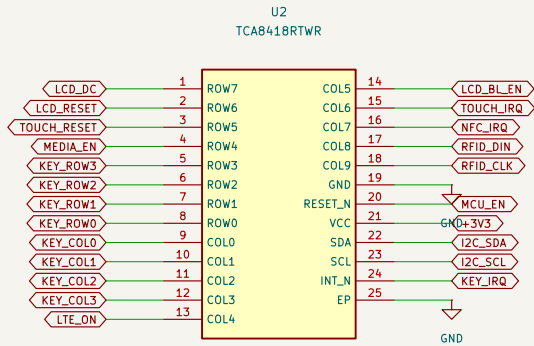
Title: ESP32 host, boot and reset

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 2/24

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/
File: keys.kicad_sch

Title: Key matrix and trackball

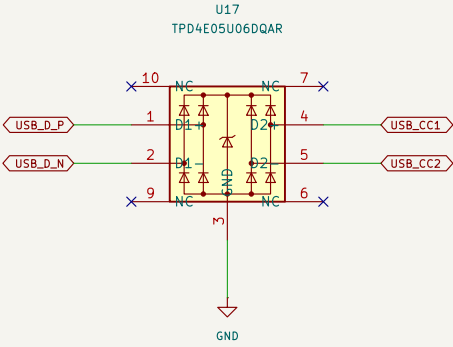
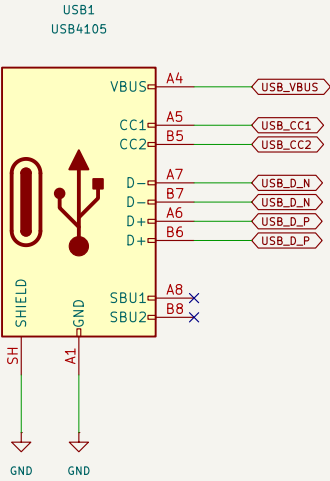
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 3/24

USB-C data and ESD

CC1/CC2 terminate at the sink controller on usb-detect. Data pair goes to the ESP32. Shield/ground return needs layout review.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/
File: usb.kicad_sch

Title: USB-C data and ESD

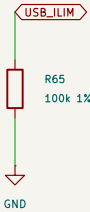
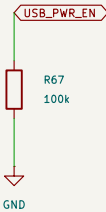
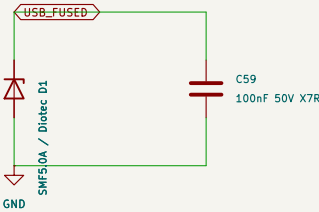
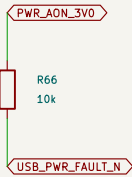
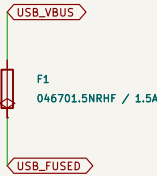
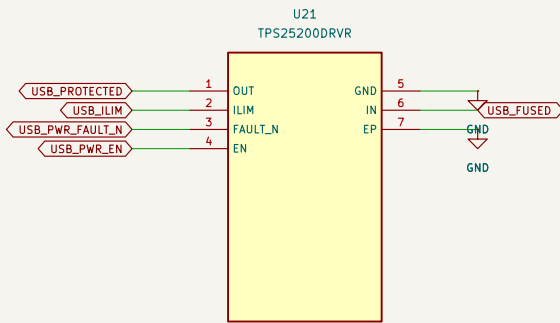
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Date: 2026-09-24

Rev: A – engineering draft
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Fused USB input and eFuse

TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

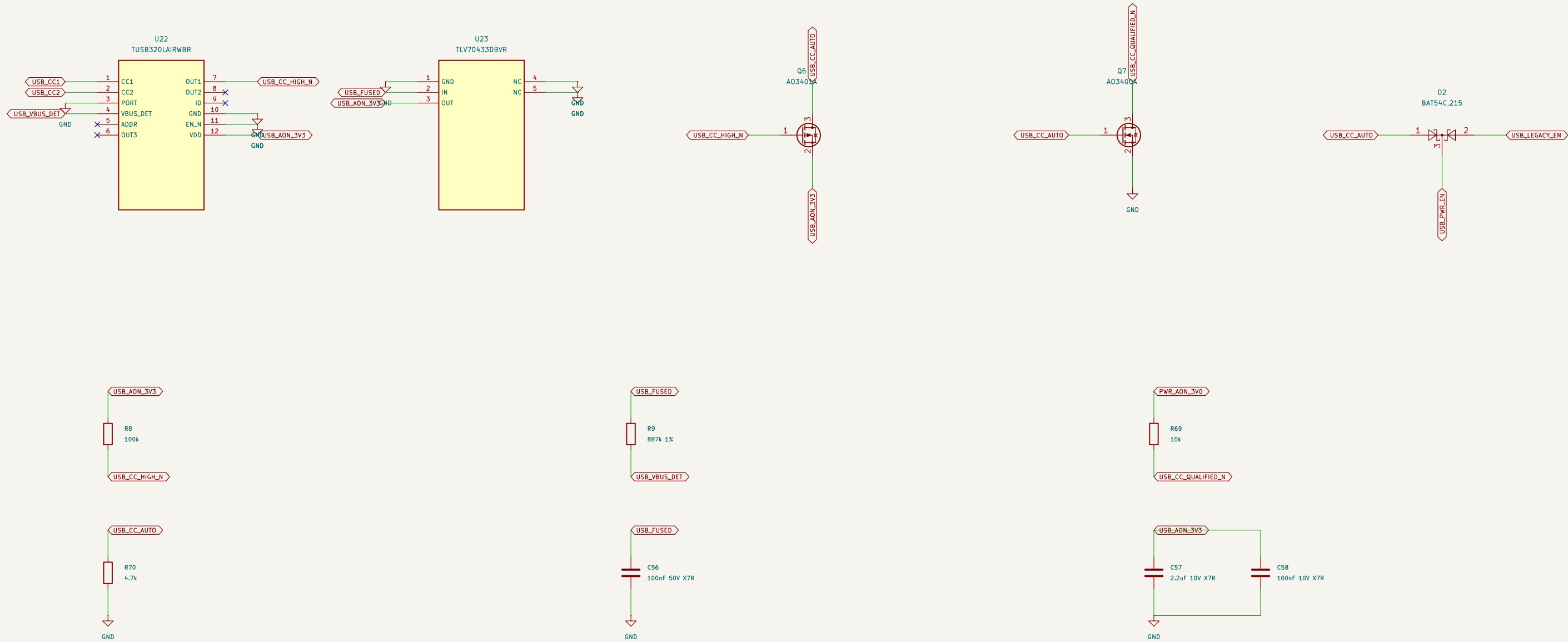
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Size: A3 Date: 2026-09-24
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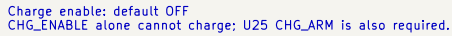
USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.



Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

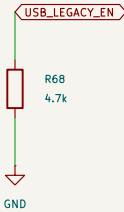
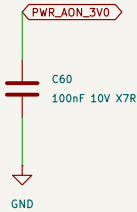
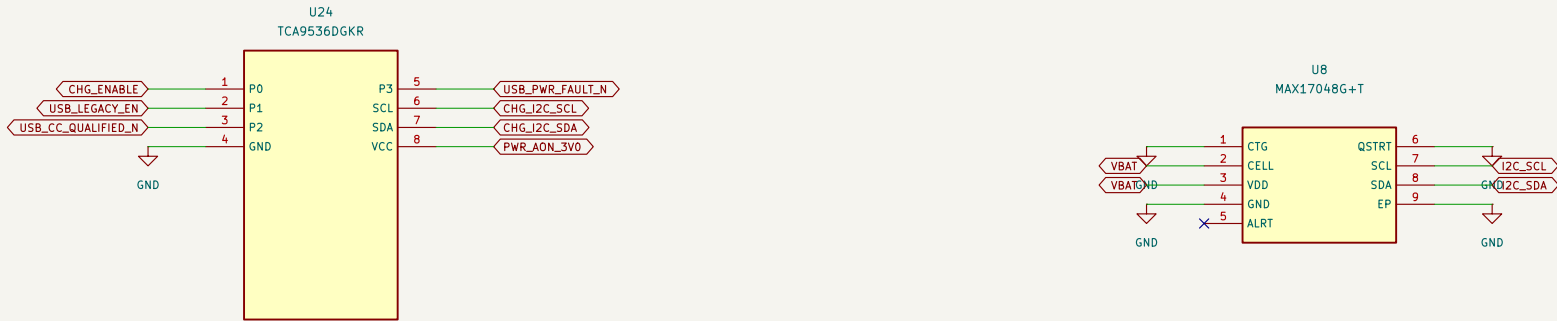
Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Size: A3	Date: 2026-09-24	Rev: A – engineering draft
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Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

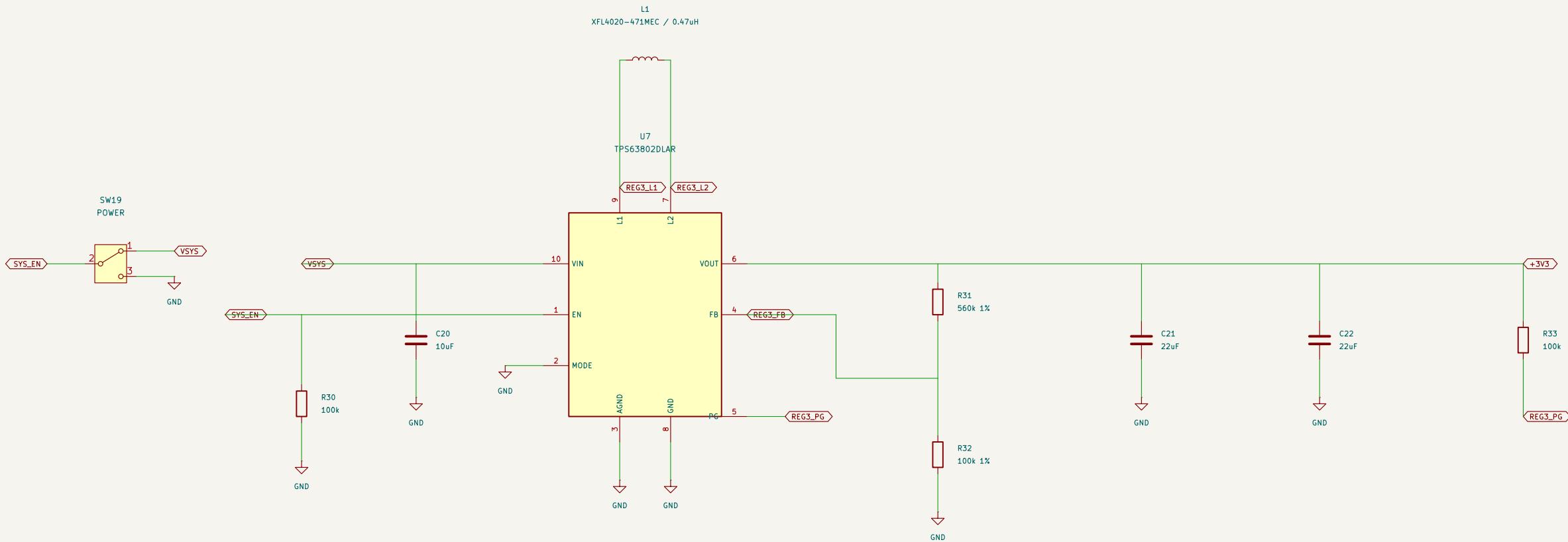
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 8/24

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/
File: reg3.kicad_sch

Title: Main 3.3 V buck–boost supply

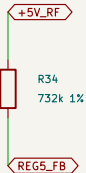
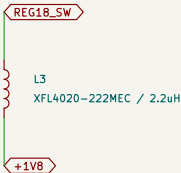
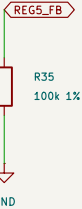
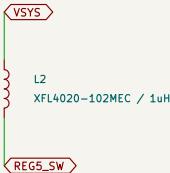
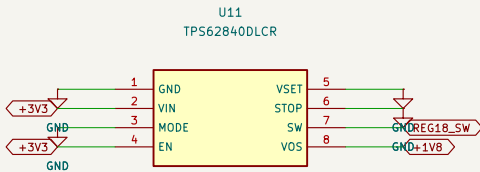
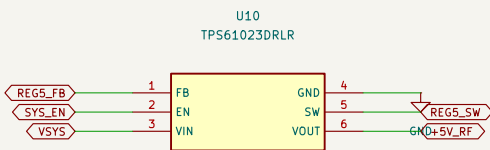
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KiCad E.D.A. 10.0.6

Date: 2026–09–24

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Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

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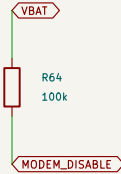
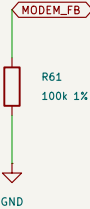
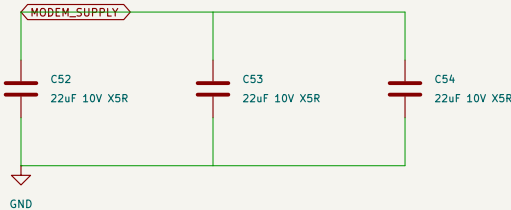
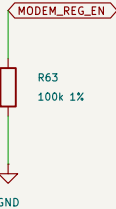
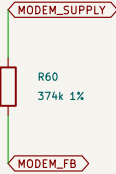
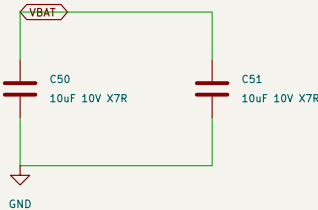
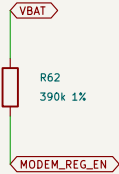
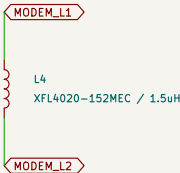
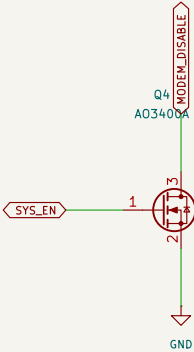
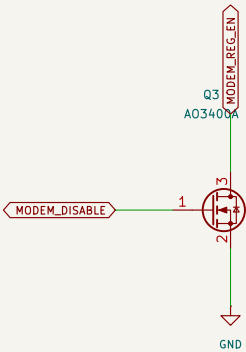
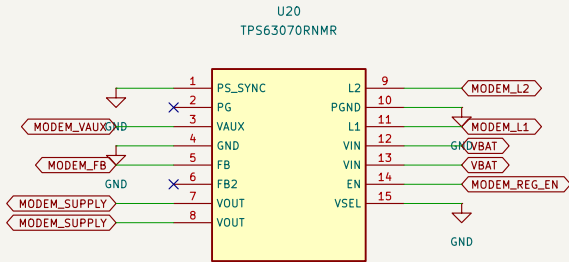
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

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Id: 10/24

Regulated modem supply and enable

Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication

Sheet: /Regulated modem supply and enable/
File: modem-power.kicad_sch

Title: Regulated modem supply and enable

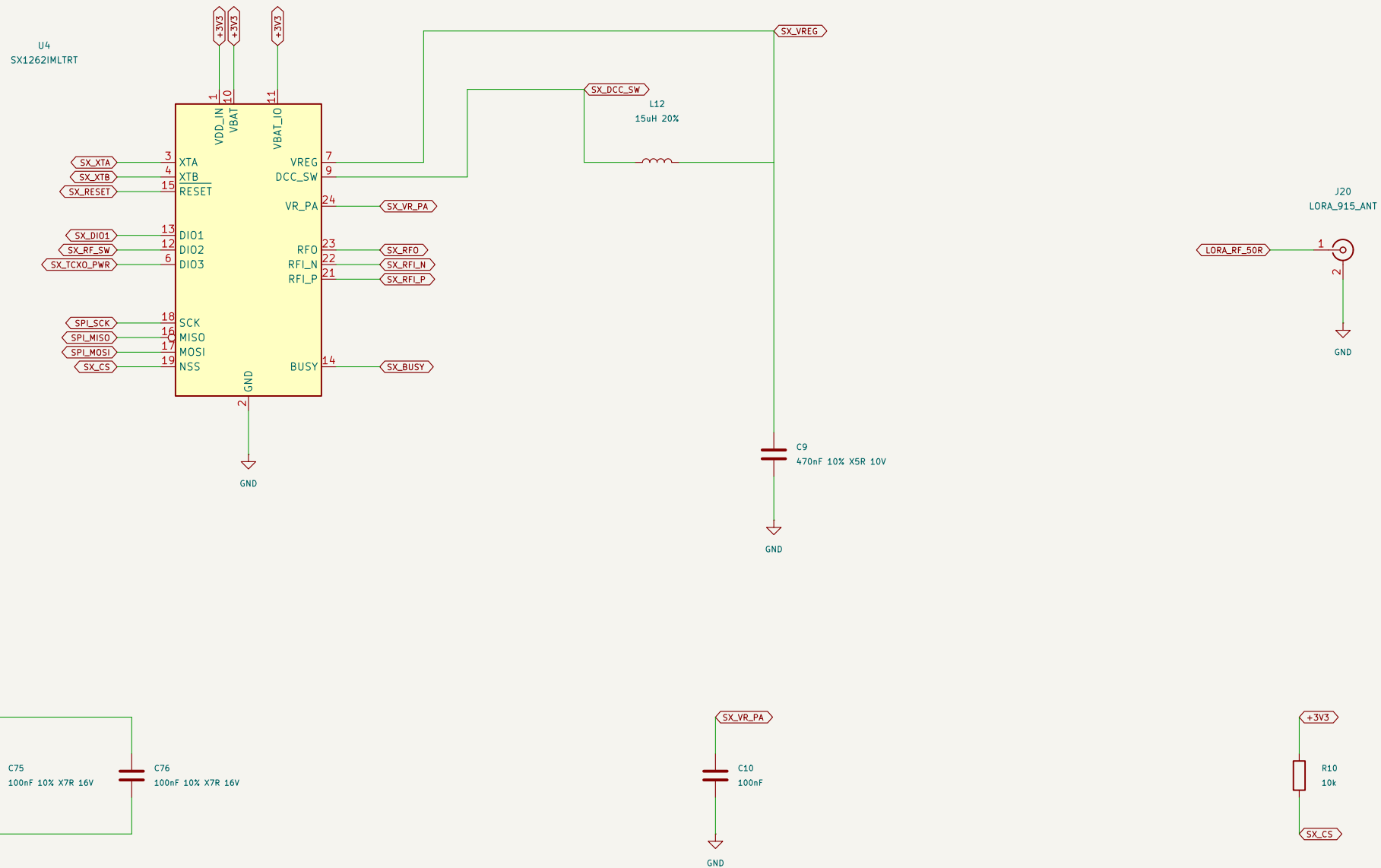
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 11/24

SX1262 LoRa radio

DC-DC support captured from Semtech DS rev1.2 Fig14-2 and Table5-3. Clock and RF frontend remain incomplete. Do not power.



C8: VBAT pin10. C75: VDD_IN pin1. C76: VBAT_IO pin11.
All use the same +3V3 rail; place each close to its pin.
C9 replaces the incorrect 100nF with Semtech reference 470nF.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
TCXO, PA feed, matching/filter and RF switch are still missing.
C10 PA bypass and J20 antenna connection remain provisional.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/
File: radios.kicad_sch

Title: SX1262 LoRa radio

Size: A3 Date: 2026-09-24

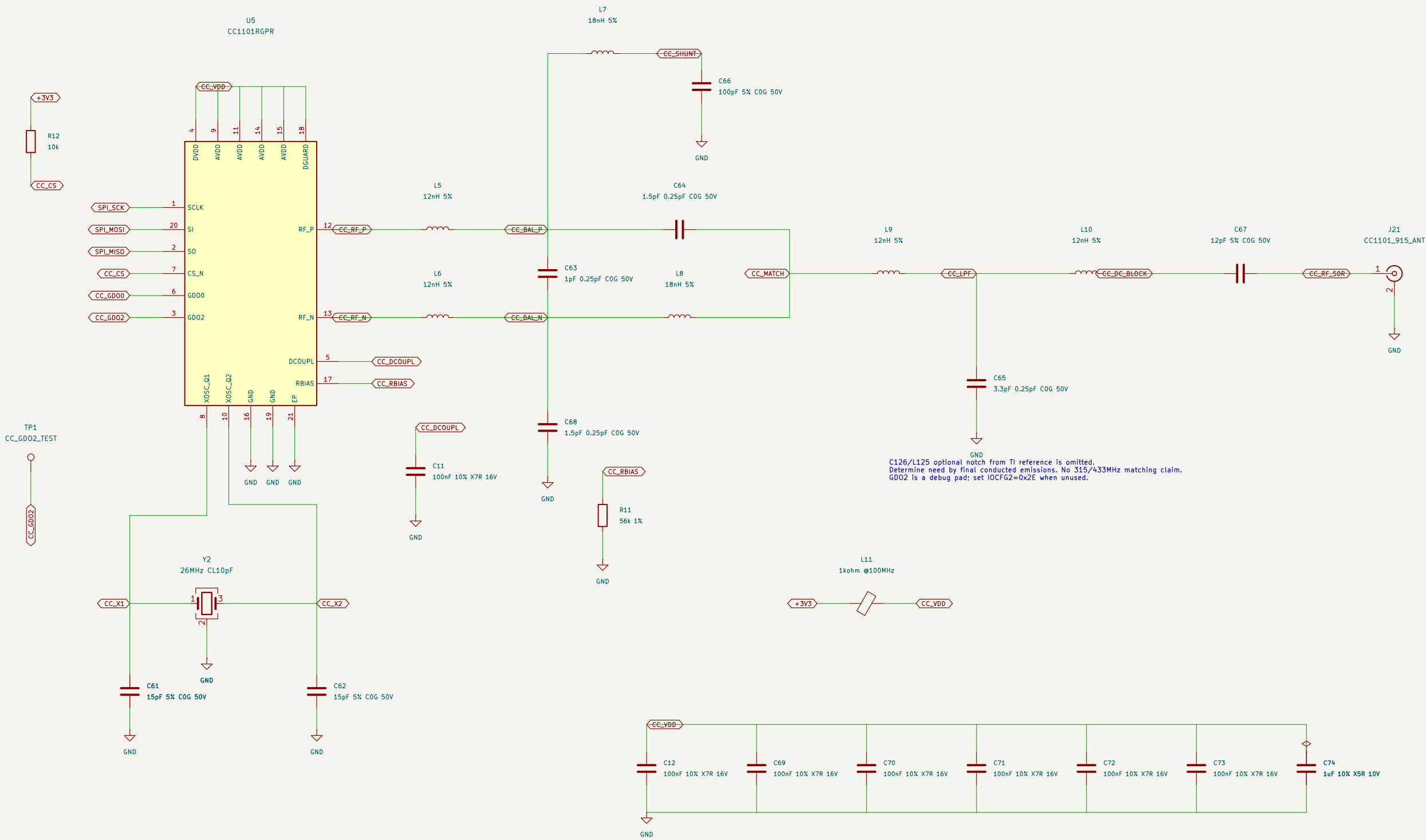
KiCad E.D.A. 10.0.6

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CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWS061l Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

C126/L125 optional notch from TI reference is omitted.
Determine need by final conducted emissions. No 315/433MHz matching claim.
GD02 is a debug pad; set IOCFG2=0x2E when unused.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/
File: cc-radio.kicad_sch

Title: CC1101 sub-GHz radio

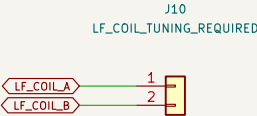
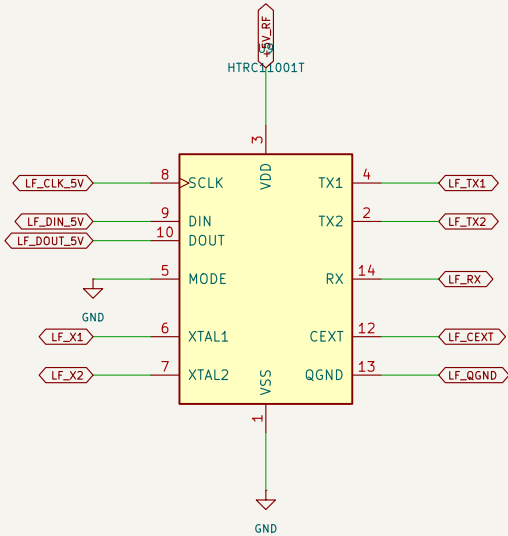
Size: A3 Date: 2026-09-24

KiCad E.D.A. 10.0.6

Rev: A – engineering draft

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

Title: 125 kHz reader analog circuit

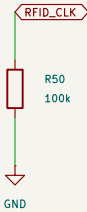
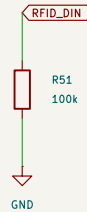
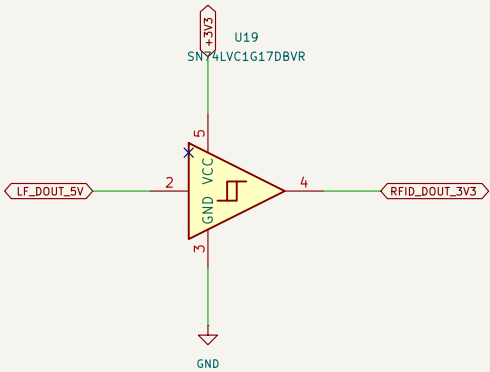
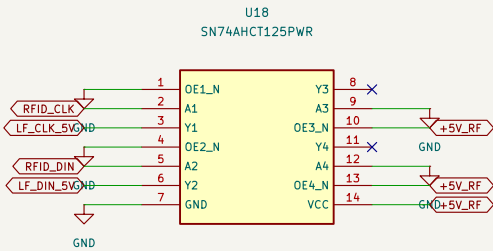
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

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Id: 14/24

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

Title: LF reader logic translation

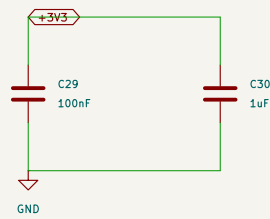
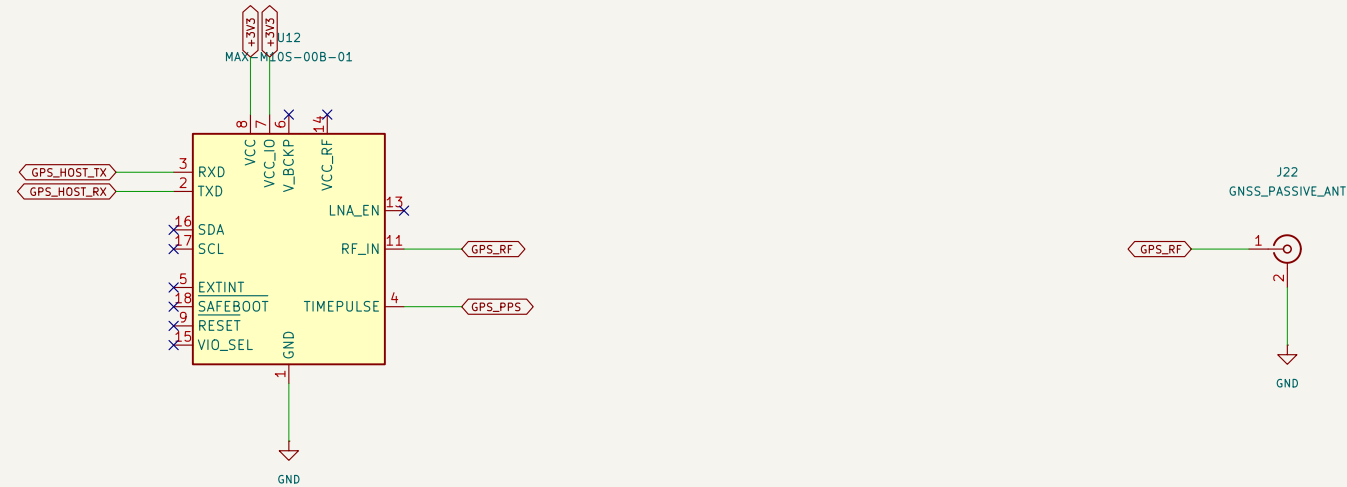
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 15/24

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

Title: GNSS receiver and passive antenna

Size: A3

Date: 2026-09-24

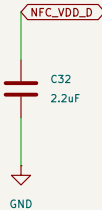
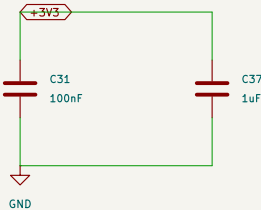
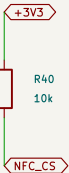
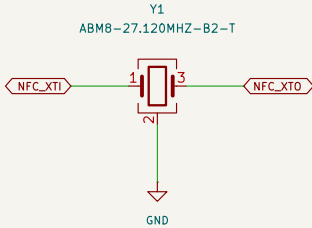
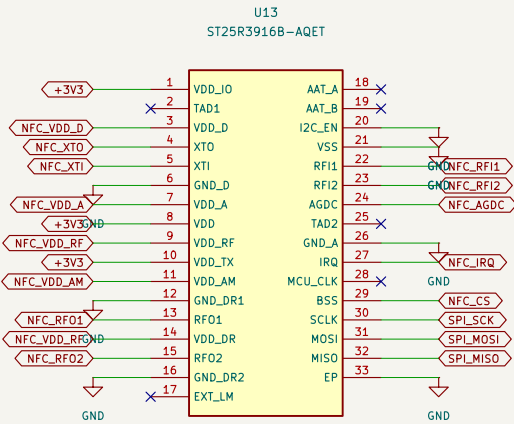
Rev: A – engineering draft

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Id: 16/24

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

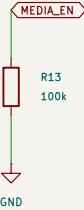
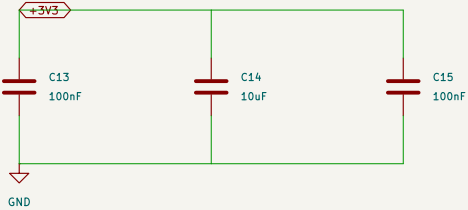
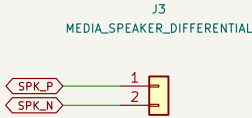
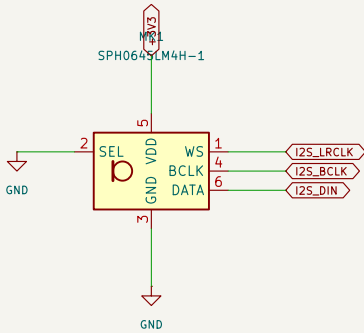
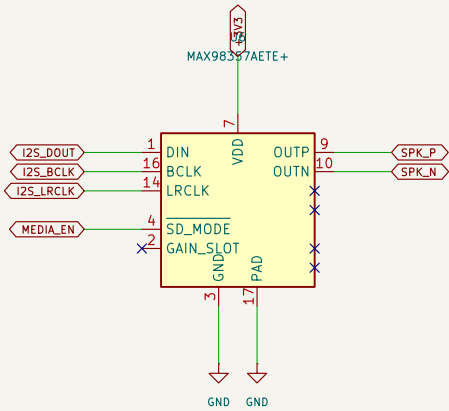
Title: NFC reader, oscillator and local supplies

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 17/24

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

Size: A3 Date: 2026-09-24

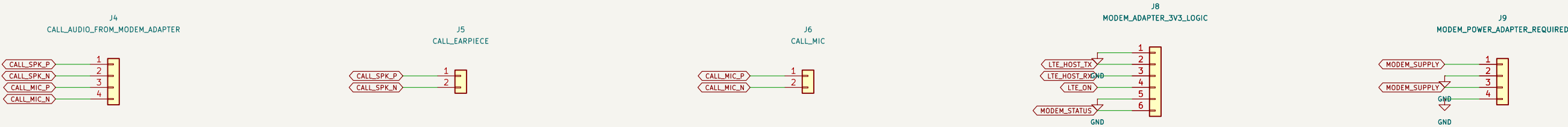
KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 18/24

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

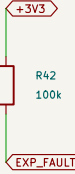
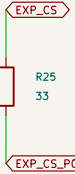
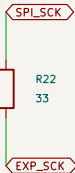
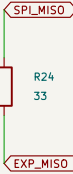
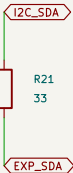
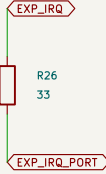
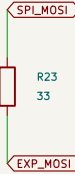
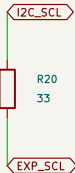
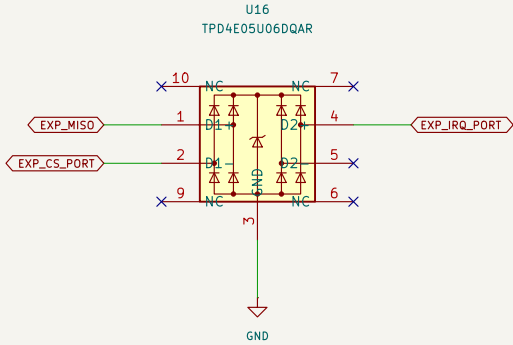
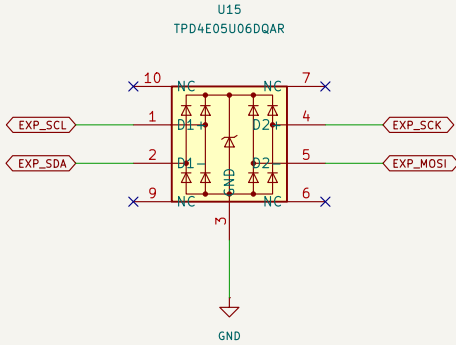
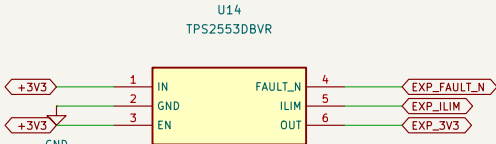
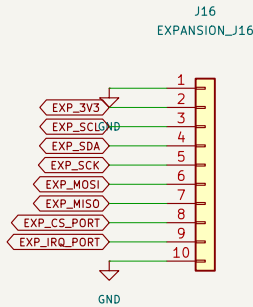
Sheet: /Cellular adapter and call–audio harnesses/
File: interfaces.kicad_sch

Title: Cellular adapter and call–audio harnesses

Size: A3	Date: 2026–09–24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 19/24

Protected external module port

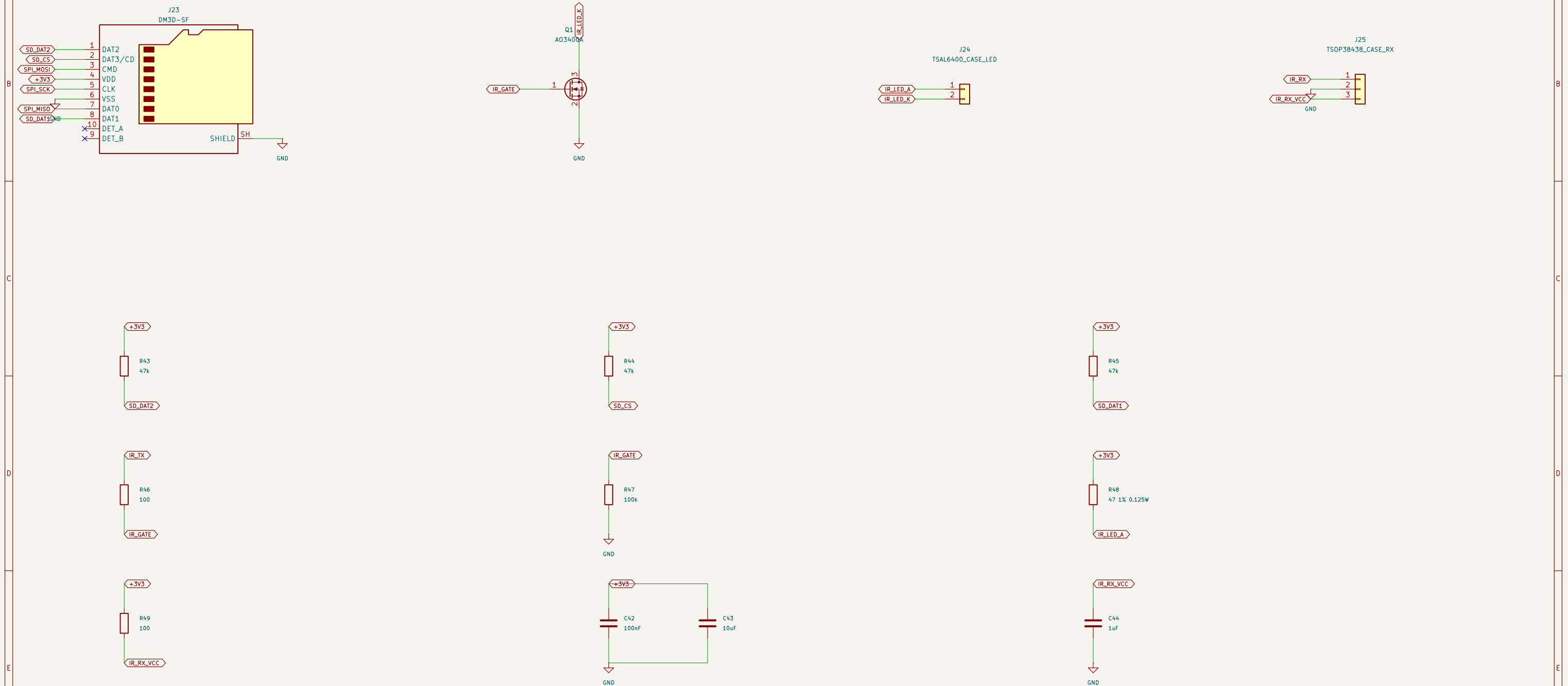
J16 pin order retained. Switch accessories with power off. TPS2553 limit and ESD path require qualification.



Circuit review in progress – not released for fabrication		
Sheet: /Protected external module port/ File: protection.kicad_sch		
Title: Protected external module port		
Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 20/24

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3

Date: 2026-09-24

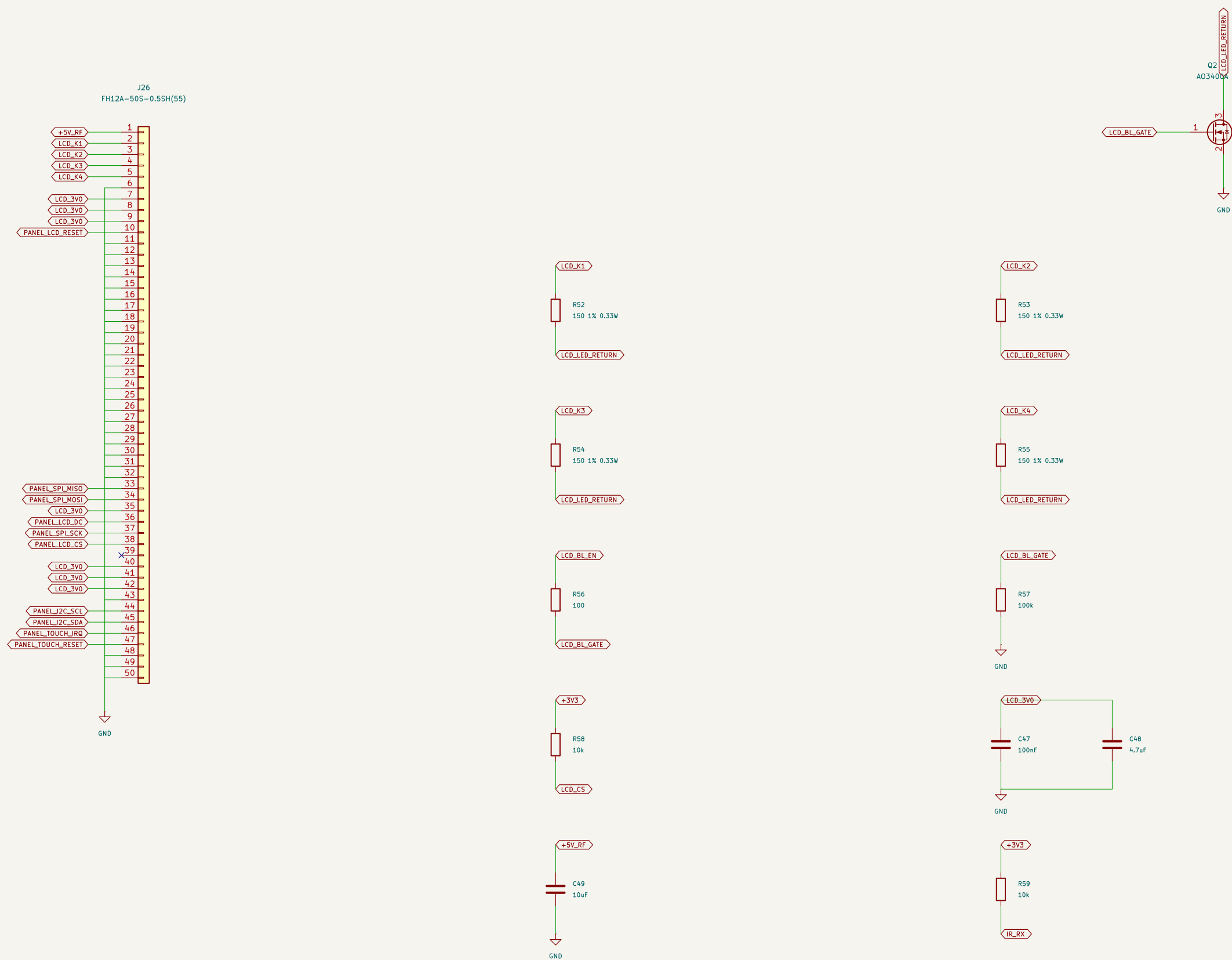
Rev: A – engineering draft

KiCad E.D.A. 10.0.6

Id: 21/24

External display, touch and backlight

Dedicated LCD_3V0 supply and translated signals on display-interface. Backlight remains on +5V_RF. Purchased CTP and timing qualification pending.



Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

Title: External display, touch and backlight

Size: A3

Date: 2026-09-24

Rev: A – engineering draft

Id: 22/24

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

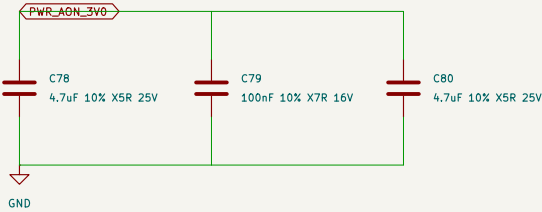
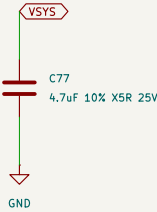
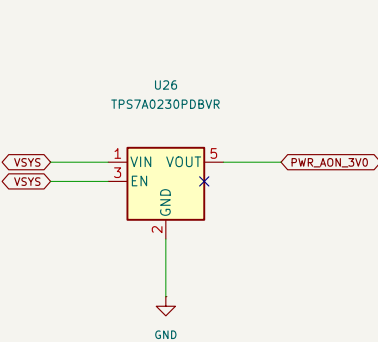
B

C

D

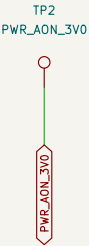
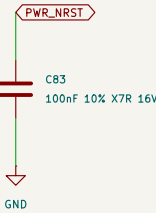
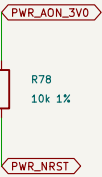
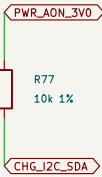
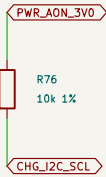
E

F



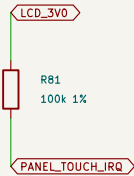
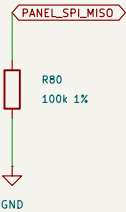
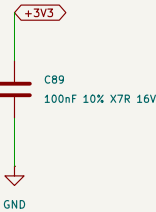
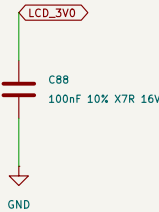
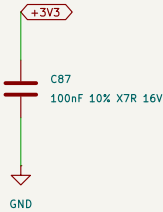
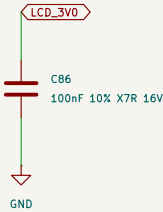
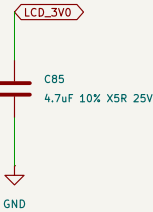
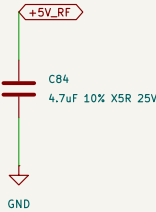
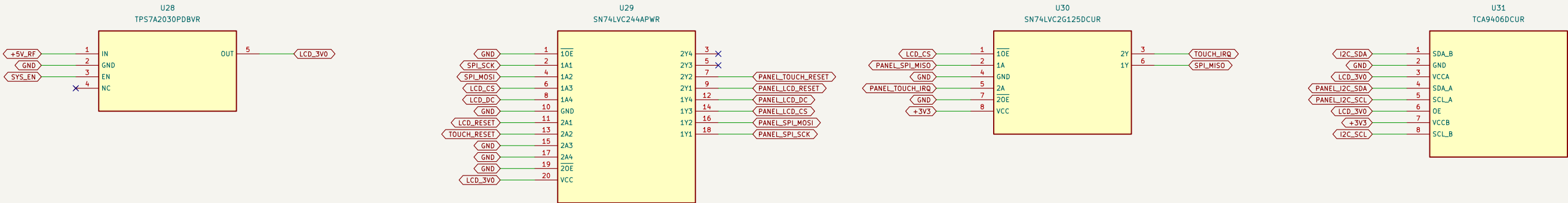
PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.

I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.



Display supply and signal translation

Interface routed and continuity checked. Qualify loads, timing, startup and case cable. Do not fabricate.



U29: 3.3 V tolerant inputs; outputs use LCD_3V0.
U30: MISO OE follows host LCD_CS; IRQ buffer always enabled.
U31: internal pull-ups; verify bus capacitance and touch option.
U28: 5 V input provides headroom; qualify load and heat.

Circuit review in progress – not released for fabrication

Sheet: /Display supply and translation/
File: display-interface.kicad_sch

Title: Display supply and signal translation

Size: A3 Date: 2026-09-24

KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 24/24